

JISHNU ROY

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PROFESSIONAL SUMMARY

OCI-certified Full-Stack Systems, DevOps, and Data Science Engineer with an academic background in Electronics and Communication Engineering. Expert in architecting distributed backend services, optimizing multi-agent generative AI pipelines, and implementing high-availability cloud infrastructure. Proven track record across software engineering lifecycles, ranging from sandboxed hardware-in-the-loop (HIL) automation to resilient REST API deployment and production-grade ML/DL modeling.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL (PostgreSQL, MySQL), JavaScript (Node.js/Express.js, React.js), MATLAB, Shell Scripting, HTML/CSS
- **Backend & Data Engineering:** FastAPI, RESTful APIs, Pandas, Scikit-learn, Vector DBs (Chroma, Pinecone), Big Data (MongoDB Atlas NoSQL, SnowSQL/Snowflake)
- **Artificial Intelligence & Systems:** GenAI Orchestration, LangChain, Large Language Models (Gemini 2.5 Flash, Qwen2.5-Coder), Ollama SLM serving, PyTorch, TensorFlow, OpenCV, YOLOv5 Computer Vision
- **Cloud & DevOps Infrastructure:** AWS (EC2, S3, RDS Aurora, IAM), Oracle Cloud Infrastructure (OCI), CI/CD Automation (GitLab CI, Jenkins, GitHub Actions), Docker Containerization, Nginx Reverse Proxies
- **Domain & Engineering Methodologies:** Embedded Systems Integration, Edge Computing (Raspberry Pi 4B), Hardware-in-the-Loop (HIL) Automation, Object-Oriented Programming (OOP), Normalized ERD Modeling, Role-Based Access Control (RBAC), Agile SDLC, Jira

EXPERIENCE

Radiant Research and Academics Pvt. Ltd.

Kolkata, WB

Backend DevOps Cloud Engineering Intern

Jan 2026 – Present

- Architected robust, secure backend microservices on AWS EC2, managing normalized ERD data modeling and high-volume relational querying using Aurora RDS (PostgreSQL) to power clinical data workflows.
- Engineered automated CI/CD deployment pipelines utilizing Git, GitLab CI, and Jenkins to perform automated database migrations and zero-downtime Nginx reverse proxy configurations.
- Built secure, stateful multi-factor authentication systems by implementing strict Role-Based Access Control (RBAC) schemas and direct Twilio API integrations.
- Streamlined operational monitoring by writing system diagnostics and logging automated scripts using Python and Shell scripting, ensuring high infrastructure up-time.

GenAI Centre of Excellence, IEM

Kolkata, WB

AI Systems Automation Engineering Intern

Dec 2025 – Mar 2026

- Co-engineered an enterprise-grade Natural Language to SQL (NL2SQL) system, connecting NLP semantic outputs to live relational databases to let non-technical stakeholders run structured database queries.
- Orchestrated multi-model agent architectures integrating Gemini 2.5 Flash Cloud APIs and local deep learning SLMs (Qwen2.5-Coder) served via Ollama to maximize inference speeds.
- Built declarative automation pipelines utilizing LangChain for intelligent data routing, low-latency API handling, and structured JSON parsing.
- Designed and integrated a Human-in-the-Loop feedback mechanism to capture runtime prompt refinements and iteratively validate backend data integrity.

IEDC, IEM

Kolkata, WB

Undergraduate Research Trainee (Embedded Systems & Deep Learning)

June 2025 – Nov 2025

- Developed an automated edge-based Virtual Barricading System featuring real-time Facial Recognition and Object Detection using optimized YOLOv5 and OpenCV vision structures.
- Deployed resource-constrained PyTorch machine learning models onto embedded hardware target layers (Raspberry Pi 4B) to minimize system processing latency.
- Executed comprehensive hardware-in-the-loop (HIL) performance metrics analysis to stabilize real-time edge streaming pipelines and eradicate hardware data delay loops.

PROJECTS

EduAi: High-Availability Multi-Agent Assessment Platform

Cloud & AI Orchestration

- Designed a full-stack, cooperative multi-agent platform using Next.js 15, Node.js, and Express to auto-generate and validate academic assessments against custom rubrics.
- Engineered an asynchronous multi-agent critique loop using a Generator Agent to draft items, a Critic Agent to evaluate compliance against Bloom's Taxonomy, and a Revision Agent to refine outputs.
- Coded a custom High-Availability Redis Bypass Pipeline for the BullMQ queuing engine that intercepts Redis drops to gracefully downgrade tasks inline, preserving 100% generation availability.
- Deployed sandboxed runtime code execution tools to programmatically fact-check and verify mathematical and code string outputs inside generated grading answer keys.
- Implemented real-time system visualization UI modules leveraging secure WebSocket connections to push granular, multi-agent execution logging streams to frontend clients.

AgriGuru: Real-Time Agriculture Data Ingestion Pipeline

Python, FastAPI, MongoDB Atlas, Docker

- Engineered a resilient backend engine using FastAPI to serve RESTful endpoints managing constant real-time data ingestion streams for dynamic market Mandi price tracking.
- Maintained unstructured big data layouts using MongoDB Atlas to securely structure, modify, and stream high-throughput agricultural edge sensor analytical arrays.
- Integrated predictive data components (CNN and Random Forest models) achieving up to 98% accuracy thresholds for ongoing analytical model evaluation tasks.

Custom UAV Control System Simulation

MATLAB Simulink, Embedded Control Logic

- Developed a full-scale digital Unmanned Aerial Vehicle (UAV) environment using MATLAB Simulink to construct embedded system control feedback loops tracking altitude regulation metrics.
- Programmed automated sensor fault-recovery logic to evaluate physical hardware responses against simulated hardware-in-the-loop sequences.

Customer Churn Statistical Analysis Pipeline

Python, Pandas, Scikit-learn

- Built automated Python pipelines utilizing Pandas to filter, clean, and run Exploratory Data Analysis (EDA) profiles over massive customer datasets to locate leading churn indicators.
- Trained a predictive Logistic Regression classification model that attained a 95% testing accuracy score.

EDUCATION

Institute of Engineering and Management

Kolkata, WB

B.Tech in Electronics and Communication Engineering (SGPA / CGPA: 8.68)

2023 – 2027

Burdwan Municipal High School

Burdwan, WB

Higher Secondary (Class XII) – WBCHSE (Score: 93%)

2019 – 2021

St. Xavier's School

Burdwan, WB

ICSE (Class X) – CISCE (Score: 94%)

2019

CERTIFICATIONS & AWARDS

- **1st Prize Winner:** FrostHacks 2026 (Winner out of multi-team pool under Sustainability and Green Earth Track) Mar 2026
- **Global Rank 6:** Infosys Global Hackathon (Selected out of international pool into Top 30 Finals teams) July 2025
- **Oracle Certified Professional:** OCI 2025 Data Science Professional (1Z0-1110-25) Nov 2025
- **Oracle Certified Professional:** OCI 2025 Generative AI Professional (1Z0-1127-25) Nov 2025
- **AWS Certified Platform Competency:** AWS Cloud Practitioner Essentials (Coursera Validation) Aug 2025